

Effect of psilocybin on depressive and anxiety symptoms in working-age adults: a systematic review and meta-analysis of randomized controlled trials

Miguel Ángel Trejo-Rodríguez, Erik González-Ballesteros, Sandra Martínez-Robles, Andrea Estefanía Aranda Rosas, Leandro Jesús Lozano García

REVIEW TITLE AND BASIC DETAILS

Review title

Effect of psilocybin on depressive and anxiety symptoms in working-age adults: a systematic review and meta-analysis of randomized controlled trials

Review type

No preview available

Condition or domain being studied

Major depressive disorder; Anxiety Disorder; Anxiety

Primary depressive disorders (including major depressive disorder and treatment-resistant depression) and primary anxiety disorders (including generalized anxiety disorder) in working-age adults (18 to 65 years). Depression or anxiety secondary to severe or terminal somatic illness (such as cancer, Parkinson disease, or dementia) is outside the scope of this review, as are conditions studied without separately reported depression or anxiety outcomes (such as alcohol use disorder or eating disorders).

Rationale for the review

Psilocybin has moved rapidly from early-phase exploration to phase 2 and phase 3 randomized controlled trials for depressive and anxiety disorders, including landmark single-dose trials in treatment-resistant depression. Existing systematic reviews and meta-analyses predominantly pool trials conducted before this recent expansion, combine heterogeneous populations (including patients with life-threatening somatic illness and existential distress), and mix psilocybin with other psychedelic compounds, which limits the clinical interpretability of pooled estimates for patients with primary depressive or anxiety disorders. This review addresses these limitations by restricting eligibility to randomized controlled trials published from 2022 to 2026 in working-age adults (18 to 65 years) with primary depressive and/or anxiety disorders, excluding depression or anxiety secondary to severe somatic illness. Pre-specified moderator analyses (dose regimen, presence of assisted psychotherapy, diagnostic population, comparator type) and sensitivity analyses (risk of bias, preprint status) will examine sources of heterogeneity. The review will provide an updated, population-specific estimate of the efficacy and safety of psilocybin, with certainty of evidence rated using GRADE, to inform clinical research and regulatory discussion during a period of accelerated trial activity.

Review objectives

Primary question: In adults aged 18 to 65 years with a primary depressive and/or anxiety disorder, does psilocybin treatment, compared with placebo or usual or active treatments, improve depressive symptom severity measured on validated scales?

Secondary questions: (1) Does psilocybin improve anxiety symptom severity, response, remission, functioning, and quality of life compared with the same comparators? (2) What is the frequency and profile of adverse events associated with psilocybin in this population? (3) Do treatment effects vary by dose regimen, presence of assisted psychotherapy, diagnostic population (depressive versus anxiety disorder), or comparator type? (4) Are pooled estimates robust to the exclusion of trials at high risk of bias and of preprints?

Keywords

Psilocybin; Psychedelic-assisted therapy; Major depressive disorder; Treatment-resistant depression; Generalized anxiety disorder; Randomized controlled trials; Meta-analysis

Country

Mexico

ELIGIBILITY CRITERIA

Population*Included*

Adults aged 18 to 65 years (working age) with a primary diagnosis of a depressive disorder (including major depressive disorder and treatment-resistant depression) and/or an anxiety disorder (including generalized anxiety disorder), established using recognized diagnostic criteria (for example DSM-5 or ICD-10/11) or a validated structured interview. Participants maintaining their daily routine and occupational functioning, without need for palliative care. Trials enrolling mixed diagnostic populations are eligible if outcomes for eligible participants are reported separately or constitute the clear majority of the sample.

Excluded

Participants younger than 18 or older than 65 years. Depression or anxiety secondary to, or explicitly reactive to, severe or terminal somatic illness (including cancer, Parkinson disease, dementia, or other serious conditions with behavioural implications). Trials enrolling participants primarily for other conditions (including alcohol use disorder, other substance use disorders, or eating disorders) without separately reported depression or anxiety outcomes. Healthy volunteers. Patients receiving palliative or end-of-life care.

Intervention(s) or exposure(s)*Included*

Psychotherapy; Psilocybin

Psilocybin administered orally as the principal intervention, in single or multiple doses under a defined dosing regimen, at any dose level, with or without accompanying psychological support or assisted psychotherapy. Both synthetic psilocybin (for example COMP360) and standardized formulations are eligible. Administration must occur in a supervised clinical or research setting. The presence and type of psychological support will be recorded and examined as a pre-specified moderator.

Excluded

Interventions combining psilocybin with other psychedelic or serotonergic compounds administered as part of the same protocol (including ayahuasca, MDMA, LSD, DMT, or ketamine), where the effect of psilocybin cannot be isolated. Microdosing regimens without a defined supervised protocol. Whole-mushroom or non-standardized preparations without defined psilocybin content. Recreational, naturalistic, or unsupervised use.

**Comparator(s) or control(s)***Included*

PICO tags selected: Placebo; Usual Care; Antidepressants; Waiting list control

Placebo or placebo-like control (including niacin or very-low-dose psilocybin used as active placebo), treatment as usual (including approved antidepressants such as escitalopram), waiting list, or active pharmacological or psychotherapeutic comparators. When a trial includes several eligible comparator arms, the primary analysis will use the least active comparator; comparator type will be examined as a pre-specified moderator.

Excluded

Comparator arms consisting of other classic psychedelics at psychoactive doses (for example LSD, ayahuasca, DMT) or of ketamine or esketamine, since these comparisons address relative psychedelic efficacy rather than the effect of psilocybin against standard care.

Study design

Only randomized study types will be included.



Included

Randomized controlled trials, including phase 2, phase 3 and phase 4 trials, with parallel-group or crossover designs (crossover trials will contribute first-period data where carry-over is plausible given the persistent effects of psilocybin). Trials must report depression and/or anxiety severity on validated scales (for example MADRS, HAM-D, PHQ-9, BDI, HAM-A, GAD-7, STAI, BAI). Peer-reviewed publications and preprints under review (retrieved through Europe PMC, including medRxiv) published from 2022 to 2026 are eligible; preprints that reach journal publication before data extraction will be replaced by the peer-reviewed version. No language restrictions.

Excluded

Phase 1 trials in healthy participants; non-randomized studies, observational studies, case reports and case series; systematic reviews and meta-analyses; animal and in vitro studies; conference abstracts without sufficient outcome data; reports whose full text cannot be retrieved after institutional access and author contact (documented as reports not retrieved in the PRISMA flow diagram).

Context

Trials conducted in any clinical or research setting (inpatient, outpatient, or dedicated research units) and in any country will be eligible; no geographic or income-level restrictions apply. Psilocybin must be administered in a supervised clinical or research context under a defined dosing protocol; studies of recreational, naturalistic, or unsupervised use are excluded. Trials published from 1 January 2022 to the final search date in 2026 are eligible; this window corresponds to the recent expansion of phase 2 and phase 3 trial activity. No language restrictions apply.

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 11 August 2026). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Therapeutic Applications of Psilocybin for Primary and Secondary Depression: A Systematic Review of Randomized Controlled Trials [published 8 May 2026] [CRD420261391152]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in population, the review looks at additional or different outcomes, the review will be more up to date, the review uses improved methods*. The authors also noted: "*This review shares psilocybin as intervention, RCT-only eligibility, and adult primary major depressive disorder as one target population. Our review differs in four substantive respects. First, that review's central aim is the comparison between primary depression and depression secondary to life-threatening illness; our review excludes secondary depression entirely and is restricted to primary diagnoses in working-age adults (18 to 65 years). Second, our review includes primary anxiety disorders as a co-primary population, which that review does not address. Third, our review is restricted to trials published from 2022 to 2026 and incorporates preprints through a pre-specified complementary source (Europe PMC/medRxiv), targeting the recent phase 2/3 expansion. Fourth, our review conducts a quantitative meta-analysis with pre-specified moderators (dose regimen, assisted psychotherapy, diagnostic population, comparator type), sensitivity analyses and GRADE rating, whereas that registration describes a systematic review without meta-analytic synthesis. We will proceed.*"
- Psilocybin for Depressive Disorders: A Systematic Review and Meta-Analysis of Randomized Controlled Trials [published 5 April 2026] [CRD420261361083]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in population, the review looks at additional or different outcomes, the review will be more up to date*. The authors also noted: "*This registration is the closest to ours: it shares psilocybin as intervention, RCT-only eligibility, and a random-effects meta-analytic synthesis. Our review differs in scope rather than rigour. First, population: that review is restricted to major depressive disorder and treatment-resistant depression, whereas ours additionally includes primary anxiety disorders as a co-primary population, restricts eligibility to working-age adults (18 to 65 years), and explicitly excludes depression or anxiety secondary to severe somatic illness. Second, outcomes: we will meta-analyse anxiety symptom severity on validated scales, together*

with response, remission, functioning and quality of life; that review addresses depressive outcomes, durability and specific safety signals. Third, currency and sources: our search will be executed after registration approval with a later cut-off, restricted to the 2022-2026 trial expansion window, and incorporates preprints through a pre-specified complementary source (Europe PMC/medRxiv). We additionally pre-specify assisted psychotherapy as a moderator and will rate certainty of evidence with GRADE. The reviews are complementary; we will proceed."

- Therapeutic and Subjective Mediators of Psilocybin Treatment for Anxiety and Depression: A Systematic Review and Meta-Analysis [published 19 January 2026] [CRD420251171691]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in population, the review looks at additional or different outcomes, the review will be more up to date*. The authors also noted: "*though its title emphasises mediators, this registration's primary objective overlaps with ours: therapeutic effects of psilocybin on anxiety and depression in adults measured on validated scales. Our review differs in three respects. First, population: our eligibility is restricted to working-age adults (18 to 65 years) with primary depressive and/or anxiety disorders, explicitly excluding anxiety or depression secondary to life-threatening somatic illness, a population that mediator-focused syntheses typically include through trials in existential distress. Second, outcomes and analytic focus: that review's distinctive contribution is the synthesis of subjective-experience measures (MEQ-30, 5D-ASC, Emotional Breakthrough Inventory) as mediators of improvement, which we do not address; our review instead synthesises response, remission, functioning, quality of life and adverse events, with pre-specified moderators (dose regimen, assisted psychotherapy, diagnostic population, comparator type) and GRADE certainty rating. Third, currency: our search will be executed after registration approval, restricted to the 2022-2026 trial expansion window, and incorporates preprints through a pre-specified complementary source. The mechanistic and the effect-estimation questions are complementary; we will proceed.*"

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

This record has not been submitted.

Review timeline

Review start date: 12 August 2026. Review end date: 31 March 2027.

Date of registration in PROSPERO

This record has not been published.

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol

A full protocol has been written and uploaded to PROSPERO. The protocol may be accessed through this link <https://www.crd.york.ac.uk/PROSPEROFILES/3ef4c818d6d1a23188671dd1de96d182.pdf>.



SEARCHING AND SCREENING

Search for unpublished studies

Both published and unpublished studies will be sought.

Main sources that will be searched

The main sources to be searched are *CENTRAL - Cochrane Central Register of Controlled Trials* and *PubMed*.

Other sources that will be searched

Europe PMC (preprint filter SRC:PPR, which indexes medRxiv) as a pre-specified complementary source for preprints under review; backward citation searching of included trials and of recent systematic reviews. Trial registries are covered indirectly through CENTRAL, which incorporates ClinicalTrials.gov and WHO ICTRP records.

Search language restrictions

There are no language restrictions.

Search date restrictions

Databases will be searched for articles published from 1 January 2022, there are no search end date restrictions.

Other methods of identifying studies

Other studies will be identified by: *contacting authors or experts, reference list checking (backward citation searching) and searching trial or study registers.*

Additional information about identifying studies

Trial registries are covered through CENTRAL, which incorporates ClinicalTrials.gov and WHO ICTRP records. Preprints are retrieved through Europe PMC (SRC:PPR filter, indexing medRxiv). Authors will be contacted for unretrievable full texts and for missing outcome data.

Link to search strategy

A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

Other relevant information about searching and screening

All searches will be executed on the same day and fully documented following PRISMA-S, recording database, interface, search date and number of records retrieved. Records will be deduplicated in Rayyan or Zotero before screening, matched by DOI and by normalized title. Two reviewers will complete a calibration exercise on a pilot sample of records before formal screening begins, with eligibility criteria refined for clarity (but not scope) if needed. Screening decisions and reasons for full-text exclusion will be documented and reported. Preprints identified through Europe PMC will be checked at data extraction against journal databases; any preprint that has reached peer-reviewed publication will be replaced by the published version, and the substitution will be documented. Retracted articles will be excluded at any stage, with verification against the Retraction Watch database for all included trials before analysis.

DATA COLLECTION PROCESS

Data extraction from published articles and reports

Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will not be contacted for further information.

Study risk of bias or quality assessment

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment

Contour-enhanced funnel plots and Egger's test ($k \geq 10$ per outcome); comparison of reported outcomes against trial registry entries and published protocols to detect selective non-reporting; search-level mitigation through preprints, trial registries and no language restrictions; publication bias rated as a GRADE domain per outcome.

Certainty assessment

Certainty of evidence will be rated per outcome using the GRADE approach, considering risk of bias, inconsistency, indirectness, imprecision and publication bias. Two reviewers will apply GRADE independently, with disagreements resolved by consensus. Results will be presented in a Summary of Findings table generated with GRADEpro GDT, reporting absolute and relative effects with certainty ratings and footnoted justifications for every downgrade decision.

Certainty will be rated separately for the primary outcome (depressive symptom severity) and each secondary outcome (anxiety symptom severity, response, remission, functioning, quality of life, adverse events).

OUTCOMES TO BE ANALYSED

Main outcomes

Primary outcome: change from baseline in depressive symptom severity, measured on validated clinician-rated or self-report scales (MADRS, HAM-D, PHQ-9, BDI or equivalent), at the primary endpoint defined by each trial; where multiple time points are reported, the pre-specified hierarchy will be the trial-defined primary endpoint, then the time point closest to 6 weeks post-dose. Where trials report the same scale, mean differences (MD) will be pooled; across different scales, standardized mean differences (Hedges' g) will be used, both with 95% confidence intervals from random-effects models (REML). Clinician-rated scales will be preferred over self-report when a trial reports both.

Co-primary outcome: change from baseline in anxiety symptom severity on validated scales (HAM-A, GAD-7, STAI, BAI or equivalent), with the same time-point hierarchy and effect measures.

Time points will additionally be grouped for sensitivity analysis as acute (up to 2 weeks), short-term (3 to 8 weeks) and sustained (12 weeks or more) post-dose.

Additional outcomes

Response: proportion of participants achieving at least 50% reduction from baseline in depressive (or anxiety) symptom severity on the primary scale, at the trial-defined primary endpoint. Remission: proportion below the established scale-specific threshold (for example MADRS ≤ 10 , HAM-D ≤ 7 , GAD-7 ≤ 5). Both will be synthesized as risk ratios (RR) with 95% confidence intervals from random-effects models; risk differences will be reported where absolute effects aid interpretation.

Functioning and quality of life: change from baseline on validated instruments (for example SDS, WSAS, WHOQOL-BREF, SF-36, EQ-5D), synthesized as MD or Hedges' g following the same rules as the primary outcomes.

Safety: proportion of participants with any adverse event, serious adverse events, and adverse events of special interest (treatment-emergent suicidal ideation or behaviour; prolonged perceptual disturbances; acute hypertensive episodes), synthesized as RR where event counts allow, otherwise summarized descriptively with exact counts per arm. Zero-event handling: trials with no events in either arm will be described but not pooled; continuity corrections will not be applied by default, and sensitivity analysis with the Mantel-Haenszel method without correction will be conducted for rare events.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

Random-effects meta-analysis will be conducted for each outcome with at least two contributing comparisons, using REML estimation of between-study variance with the Hartung-Knapp-Sidak-Jonkman adjustment for confidence intervals. Effect measures: MD (same scale) or Hedges' g (different scales) for continuous outcomes; RR for dichotomous outcomes. Heterogeneity will be quantified with I^2 , τ^2 and Cochran's Q, and 95% prediction intervals will be reported for all pooled estimates. Multi-arm trials will contribute the pre-specified primary comparison (least active comparator); if two arms of the same trial are included in one analysis, the shared control group will be split to avoid double counting. Crossover trials will contribute first-period data. Pre-specified moderator analyses (subgroup analysis or random-effects meta-regression as appropriate, conducted when at least 10 comparisons are available): dose regimen, presence of assisted psychotherapy, diagnostic population (depressive versus anxiety disorder), comparator type. Pre-specified sensitivity analyses: exclusion of trials at high risk of bias; exclusion of preprints; time-point grouping (acute, short-term, sustained post-dose); leave-one-out analysis. Any post-hoc analysis will be explicitly labelled as such. Analyses will be performed in R (metafor and meta packages); analysis scripts and derived datasets will be released in the project's public repository. If quantitative synthesis is not feasible for an outcome (fewer than two comparisons or

extreme clinical or methodological heterogeneity), results will be reported narratively following SWiM (Synthesis Without Meta-analysis) guidelines, with structured tables and effect direction plots.

LIVING REVIEW METHODS

Living systematic review methods

No preview available

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage	Started	Completed
Pilot work	✓	
Formal searching/study identification		
Screening search results against inclusion criteria		
Data extraction or receipt of IPD		
Risk of bias/quality assessment		
Data synthesis		

Review status

The review is currently planned or ongoing.

Publication of review results

Results of the review will be published in English.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

Professor Miguel Ángel Trejo-Rodríguez (review guarantor and contact) ORCID: 0000-0001-8251-6665. UNAM. Mexico.

No conflict of interest declared.

Professor Erik González-Ballesteros. UNAM. Mexico.

No conflict of interest declared.

Dr Sandra Martínez-Robles. UNAM. Mexico.

No conflict of interest declared.

Miss Andrea Estefanía Aranda Rosas. UNAM. Mexico.

No conflict of interest declared.

Mr Leandro Jesús Lozano García. UNAM. Mexico.

No conflict of interest declared.

Named contact

Professor Miguel Ángel Trejo-Rodríguez (farma_matr@cuautitlan.unam.mx). ORCID: 0000-0001-8251-6665. UNAM. Mexico.

Review affiliation

Facultad de Estudios Superiores Cuautitlán, Universidad Nacional Autónoma de México

Funding source

Review has no specific/external funding but is supported by guarantor/review team (non-commercial) institutions.

Additional information about funding

No external or specific funding. The review is supported by the authors' academic institution (UNAM) through protected academic time and institutional library access to full texts.

Peer review

There has been no peer review of this planned review.

ADDITIONAL INFORMATION

Additional information

This review is the meta-analytic arm of a hybrid study whose parallel bibliometric arm has been completed and published, together with the full protocol, at <https://doi.org/10.5281/zenodo.21893380> (GitHub: psilocybin-bibliometrics-2026). The deposit includes the PRISMA-P/PRISMA-S protocol, all search strategies, the tri-database deduplication pipeline, the derived bibliometric corpus (n = 1,921 unique documents, 2022-2026), descriptive tables and figures. The bibliometric arm maps the structure and growth of the field and does not extract clinical outcome data; no screening, extraction or synthesis of the present review had begun at the time of this registration. Analysis scripts, extraction datasets, risk of bias assessments and GRADE tables of this review will be released in the same public repository upon completion. Protocol amendments will be versioned in the repository and reflected in this record's audit trail.

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Adult; Alcoholism; Anxiety; Anxiety Disorders; Classification; Dementia; Depression; Depressive Disorder, Major; Feeding and Eating Disorders; Humans; Neoplasms; Parkinson Disease; Psilocybin; Psychotherapy; Quality of Life; Randomized Controlled Trials as Topic

PROSPERO version history

No preview available

Disclaimer

The content of this record displays the information provided by the review team. PROSPERO does not peer review registration records or endorse their content.

PROSPERO accepts and posts the information provided in good faith; responsibility for record content rests with the review team. The guarantor for this record has affirmed that the information provided is truthful and that they understand that deliberate provision of inaccurate information may be construed as scientific misconduct.

PROSPERO does not accept any liability for the content provided in this record or for its use. Readers use the information provided in this record at their own risk.

Any enquiries about the record should be referred to the named review contact